

Feed an AI-agentic beast with proper data



A hands-on workshop · 90 minutes

Jessica Eggen

01.

Welcome

Introduction

- My own background
- Who are you?
- My current work struggles
- Prompt design vs data design

Workshop goal:

Prepare your data and system architecture to enable effective LLM reasoning and deliver trustworthy, accurate results.

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Microsoft Corp. CEO **Satya Nadella** warned employees against AI overuse, urging cost-efficient model selection as rising infrastructure costs pressure Big Tech to rein in token consumption.

"I'm a tokenmaxxer too, it's addictive," Nadella said, speaking at The New York Times' "Hard Fork" podcast. "But you have to step back when the novelty wears off."

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AI • TECH

Microsoft reports are exposing AI's real cost problem: Using the tech is more expensive than paying human employees

By **Jake Angelo**
Former News Fellow

May 22, 2026, 12:56 PM ET



Microsoft CEO Satya Nadella
SVEN HOPPE/PICTURE ALLIANCE VIA GETTY IMAGES

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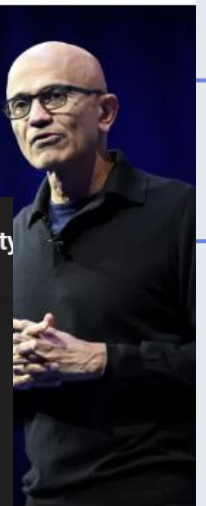
AI tokens are catching some companies by surprise — and spurring big bills



By **Ariel Rabinovitch** • Global News

Posted June 10, 2026 4:10 pm · Updated June 10, 2026 4:13 pm · 4 min read

Global News Morning Halifax



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Microsoft Corp. CEO Satya Nadella warned employees against AI overuse, urging cost-efficient model selection as rising infrastructure costs pressure Big Tech consumption.

"I'm a tokenmaxxer too, it's addictive," Nadella said at The New York Times' "Hard Fork" event. He also said AI is getting too expensive.

AI • TECH

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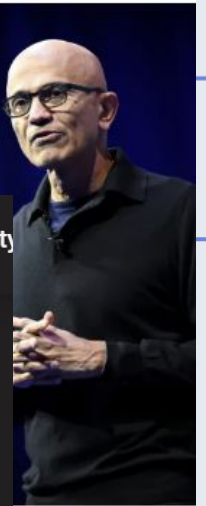
AI cost crisis hits tech giants as employee 'tokenmaxxing' backfires, sparking corporate pullback at Microsoft, Meta, and Amazon — agentic AI eats up to 1000x more tokens than standard AI

News By Jowi Morales published May 23, 2026

AI is getting too expensive.

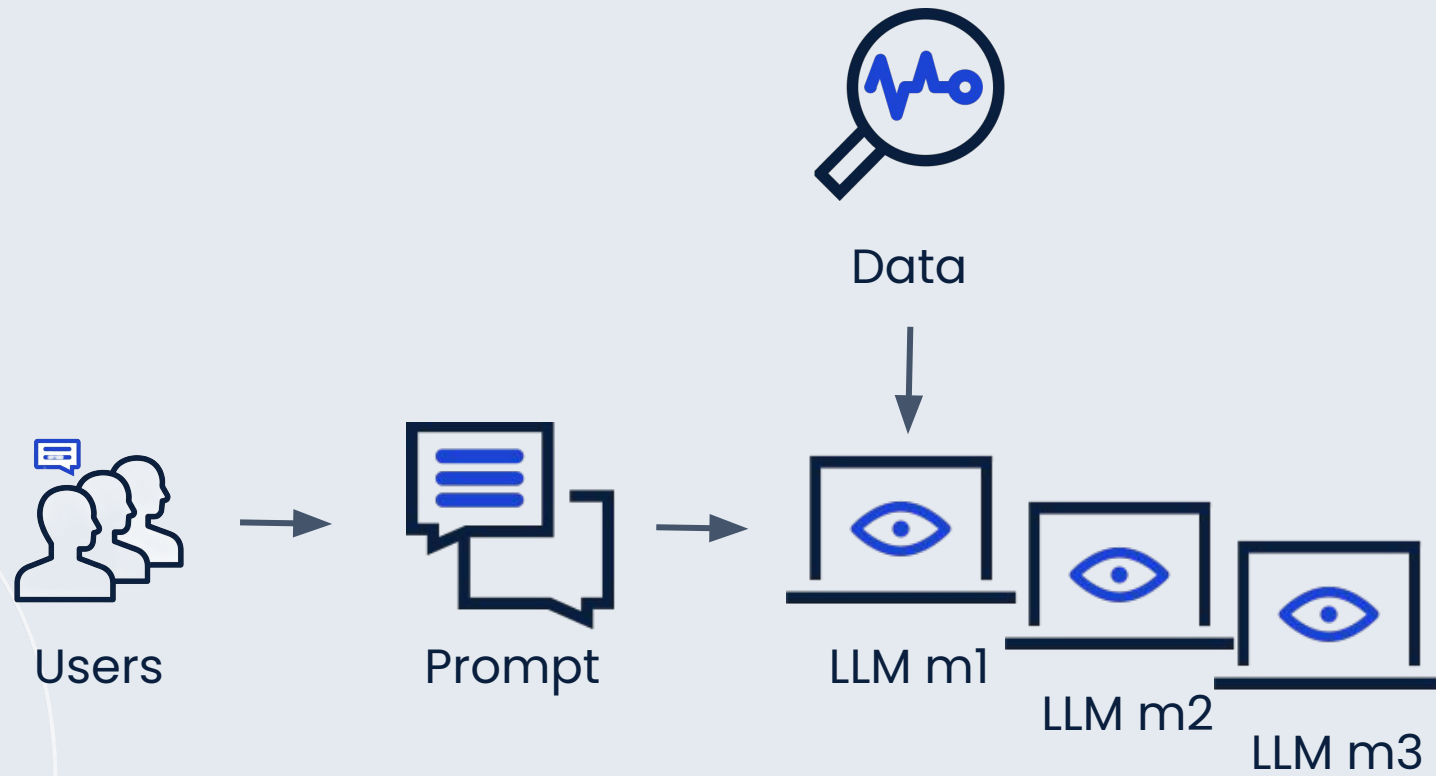
companies by surprise

3 pm · 4 min read



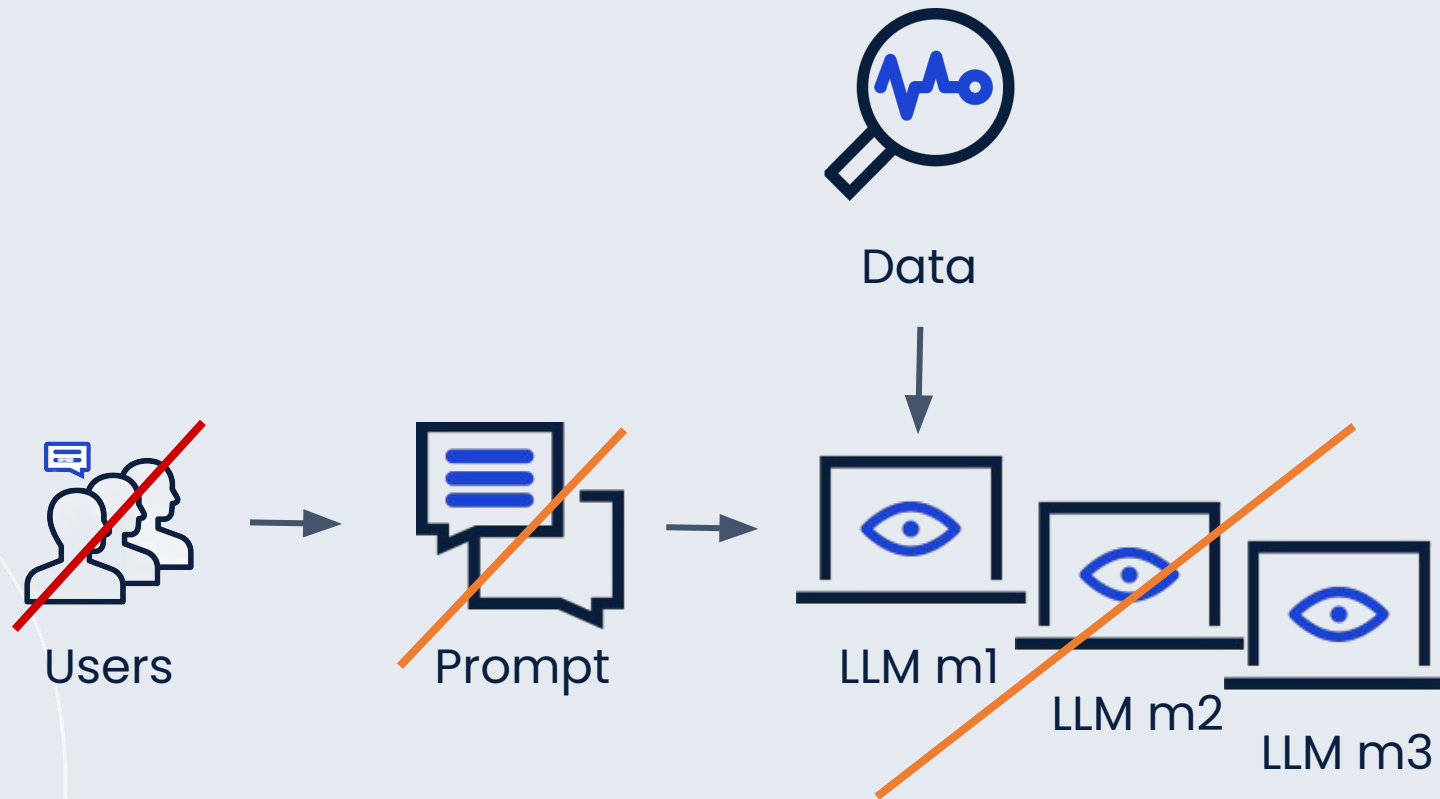
Introduction

- What can we adjust



Introduction

- What can we adjust



02.

Let's open the workbook!

Setup and exercise 1: **Scrub before you send**

Let me, or your neighbour, know if any questions pop up.

Go through the DISCUSS items with one of your neighbours!

Setup and exercise 1: scrub before you send

Everyone knows: data has to be cleaned.

Scrubbing PII is only 1 part, but there are silent NaN drops, unexpected and incorrect values, averages vs medians, abbreviations, missing units, duplicates and many more. But what is worth fixing?

There is a difference in cleaning for AI vs cleaning for a ML model or a dashboard.

Dashboard = visual correctness

ML model = statistical distribution

LLM input = semantic clarity

AI needs richer context than ML. Here's a 4-step pipeline:

1. Clean — fix errors, nulls, and inconsistencies
2. Contextualise — add table descriptions, business meaning and semantic context (what does this value mean?)
3. Chunk — split into pieces the LLM can process
4. Verify — validate output quality before sending to production

Exercise 2: Minimum viable context

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Exercise 2: Minimum viable context

Avoid

Let the LLM do the groupby

Sending raw rows and asking the LLM to aggregate is slow, expensive, and error-prone. The model may miss groups, apply wrong logic, or run out of context window. Don't make the LLM do your pandas work.

Better

Pre-aggregate to match the question

Match the granularity of your aggregation to the granularity of your question. Pre-aggregate over the dimensions of the question.

Exercise 3: Right context, right question

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Exercise 3: Right context, right question

- Irrelevant columns cost tokens and add noise.
- When LLM decides on columns it requires good column descriptions. The semantic layer is extremely important for efficient LLM use.

Exercise 4: When the LLM transforms the data

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Exercise 4: When the LLM transforms the data

- Real-life issue
- `parse_llm_json` might be irrelevant for the state-of-the-art LLM models out there, however those are more costly.
- Can create a dead-letter queue for rows that never pass
- Storing `raw_results`
This holds what the LLM said before parsing. In production, storing that alongside the parsed result means if you change your schema or validation rules later, you can revalidate old extractions without paying for new API calls

Exercise 5: Numbers they can trust

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Exercise 5: Numbers they can trust

- This is how production analytics AI works
Databricks and Tableau use text-to-SQL.
Business user → question → Tableau → writes SQL → database runs it → LLM explains the result.
- The failure in the naive approach is a cognitive task mismatch, not a knowledge gap. The model is like a brilliant analyst who can't do maths in their head, but with a calculator (the database) it is fine.
- The LLM writes SQL based on what you tell it about the table.
Where good documentation was first a nice-to-have, now it directly affects answer correctness

The AI-beast **has** been fed

Thank you for joining!

Today we covered how data flows through an LLM and explored practical data preparation techniques. The notebook has everything you need to keep practising.

Resources:

[Presentation on Github](#)

[Youtube recording](#)

[Notebook on GitHub](#)

All exercises are in the shared notebook. Fork it and keep experimenting.

Get in touch with

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